

# Jimmy Lee 李兆康

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## ACADEMIC POSITIONS

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### Postdoctoral Fellow in Economics

Global Poverty Research Lab, Northwestern University

Sept 2024 – Present

### Visiting Scholar

Center for International Development, Harvard University

Jan 2025 – June 2026

## RESEARCH FIELDS

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### Development Economics, Program Evaluation and Scaling

*Focus—Synergies between Agriculture, Rural Education, and Digital Technologies*

### Family Economics, Organizational Economics

*Focus—Intergenerational Dynamics in Rural Household and Community Decision Making*

### Behavioral Economics with applications in program evaluation, agriculture, and education

*Focus—Endogenous model uncertainty and information acquisition under cognitive limits*

## EDUCATION

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**Ph.D., Economics**, Northwestern University

2024

**M.Phil., Economics**, The Chinese University of Hong Kong

2017

**B.SSc., Economics**, The Chinese University of Hong Kong

2015

## JOB MARKET PAPER

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### “Miscategorization as an Information Friction? Activating School-Based Agricultural Extension in Liberia with Parental Engagement”

*(Funded by AFD Fund for Innovation in Development, USAID-DIV, Wellspring Foundation, and FAO)*

Schools in developing countries often have weak ties with less-educated, poorly informed parents. This paper studies the extent to which development program success hinges on strengthening these ties. In a three-year Liberian experiment across 197 schools, I evaluate school-based agricultural extension (SBAE) against control, cross-randomizing whether to engage parents via promotional videos and field days. SBAE improves targeted outcomes only when paired with engagement: students’ households adopt 26–53% more agricultural techniques, dropouts fall 4–5 percentage points per year, absenteeism falls 28%, and studying hours rise 22%. The program is more cost-effective than most extension programs in Africa and over twice as cost-effective as conditional cash transfers in raising attendance. SBAE has strong positive training effects on students that are unaltered by parental engagement. However, without parental engagement, SBAE’s diffusion mechanisms via school demonstration farms, home projects, and intra-household communication stay muted—technology adoption effects vanish, absenteeism rises 31%, studying hours fall 20%, and parents disconnect—reducing school farm visits 12% and parent–teacher meetings 39%. Parents’ avoidance of information about a new program, plus sharply reversed responses to SBAE under engagement, pose a puzzle. I discuss how standard Bayesian updating under a fixed prior, selective attention, and associative memory each fall short as standalone explanations. I propose an integrative explanation that parental engagement shifts parents’ categorization of SBAE away from the “school-garden-as-drudgery” schema.

## WORKING PAPERS

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### 1. “Technology Diffusion and Schooling as Complements: The Promise of School-Based Agricultural Extension in Africa” (joint with Chris Udry)

Efforts to raise agricultural productivity in sub-Saharan Africa are limited by substantial heterogeneity in which technologies are suitable across farms, while low perceived returns to schooling drive students to drop out. We develop a theoretical model of why school-based agricultural extension (SBAE) is suited to address this dual problem. The model extends the target-input framework with two signals of technological suitability, from a school demonstration plot and a home project, both increasing in

precision with the student's study effort. Three results follow. First, schooling and adoption become complements: the same information raises both the value of the new technology and the value of attendance. Second, schooling acquires an option value on the farm: by raising the probability of future adoption, it makes attendance optimal even where wage returns are low. Third, SBAE addresses suboptimal experimentation by decoupling learning about a technology's suitability from its use in production: because information is acquired through study effort, the returns to schooling pay part of the cost of learning. We provide experimental evidence from Liberia that SBAE cost-effectively raises both agricultural technology adoption and school attendance, and use extensions of the model to explain how SBAE's design can be strengthened.

## 2. **"Endogenous Model Uncertainty and Curiosity under Cognitive Limits"**

In standard theories of decision under model uncertainty, the weights an agent places on competing models of the world are not subject to cognitive limits. This paper models contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain: from a precise default model and a menu of latent, imprecise, and potentially more promising alternatives, an agent chooses how much attention to pay to each and how much information to acquire. The choice is disciplined by a microfounded evidence constraint that bounds motivated beliefs: the weight assigned to an alternative is capped by its likelihood relative to the default and to the other alternatives. Curiosity then arises endogenously as a drive to resolve the model uncertainty that evidence for a more promising alternative creates. The model matches regularities of neural remapping, with two results that matter for stimulating curiosity. First, contrast matters because an alternative is activated only when the evidence makes the default indefensible against it, and thresholds govern how attention shifts: gradually, abruptly, or not at all. Second, concentration of evidence matters because several alternatives may each beat the default and yet none is activated. I apply the framework to the design of information interventions that aim to draw parents' attention to a school-based agricultural extension program and to foster a conceptual understanding of pedagogy choice and learning efforts among teachers, students, and farmers.

## 3. **"Communicating Parents' Beliefs to Students as a Pivotal Component of Parental Engagement: Evidence from School-Based Agricultural Extension in Liberia"**

*(Funded by NSF, Weiss Fund, Global Poverty Research Lab, and the Buffett Institute for Global Affairs)*

When school programs require students to introduce improved techniques on household farms, is informing students and parents about the program sufficient, or is further signaling parents' support to students needed? I study this question within a randomized evaluation of school-based agricultural extension (SBAE) in Liberia, where students take school farm training into a home farming project. After parents view a promotional video of the program and students are encouraged to diffuse techniques to their parents, I randomize a nudge revealing to students their parents' optimism about their farm-management skills. Two years later, SBAE improves students' livelihoods only with the nudge: income rises 63%, on-time school fee payment 43–47%, and cost-effectiveness for attendance doubles. Analysis of mechanisms highlights a management aspect of agricultural technology adoption that can be addressed with nudges: even where parents become attentive to SBAE and adopt promoted techniques, third-party assurance is pivotal in shifting students away from assisting in subsistence farming.

## **WORK IN PROGRESS**

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*Ongoing Randomized Experiments (scaling SBAE across Liberia, Côte d'Ivoire, and Kenya in 2026-35)*

1. **"Learning to Learn: Synergizing Inquiry-based Science and Agricultural Transformation in Liberia"** (joint with Vesall Nourani, Harvard GSE; scheduled RCT within the scaling of SBAE in 2026-29)
2. **"Parental Engagement in School-based Agricultural Extension: Testing Mechanisms at Scale"** (piloted interventions in 2025-26)
3. **"Parental Engagement and the Use of Unconditional Cash Transfers"**
4. **"Directed Technical Change under Data-Driven Farming Advice: An Empirical Framework"** (grant from Weiss Fund; piloted survey method; planning RCT)

## **GRANTS AND AWARDS**

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*Total program evaluation and academic research grants secured for SBAE in Liberia: \$4.2 million*

### **Program Evaluation Grants**

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|------------------------------------------------------------------|------|
| AFD Fund for Innovation in Development (\$1.2 mil; co-PI)        | 2025 |
| J-PAL Learning for All Initiative: Scaling Award (\$222K; co-PI) | 2024 |
| USAID Development Innovation Ventures (\$395K; co-PI)            | 2022 |

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|-----------------------------------------------------------|------|
| Wellspring Foundation (\$192K; co-PI)                     | 2022 |
| AFD Fund for Innovation in Development (\$1.2 mil; co-PI) | 2021 |
| Food and Agriculture Organization (\$68K; co-PI)          | 2021 |

#### **Academic Research Awards**

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|------------------------------------------------------|------|
| J-PAL Learning for All Initiative (\$400K; co-PI)    | 2025 |
| Weiss Fund (\$5K; Lead PI)                           | 2022 |
| National Science Foundation (\$451K; co-PI)          | 2020 |
| Weiss Fund (\$29K; Lead PI)                          | 2020 |
| Global Poverty Research Lab (\$19K; Lead PI)         | 2019 |
| Buffett Institute for Global Affairs (\$5K; Lead PI) | 2019 |

#### **INVITED PRESENTATIONS**

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##### *During postdoctoral studies:*

National Chung Cheng University; NYU Africa House—Center for Technology, Economics, and Development (CTED); Mohammed VI Polytechnic University (UM6P); Tel Aviv University; Oklahoma State University; Harvard Center for International Development; Africa Food Systems Forum 2024

##### *During PhD:*

Meeting of the Society of Economics of the Household (SEHO); J-PAL—Middle East and North Africa (MENA); CGIAR—Standing Panel on Impact Assessment (SPIA); International Food Policy Research Institute (IFPRI); University of Glasgow; National Graduate Institute for Policy Studies (GRIPS, Tokyo); Development Day 2023

#### **TEACHING**

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##### **Teaching Assistant**, Northwestern University

Introduction to Microeconomics, Introduction to Macroeconomics, Intermediate Microeconomics II, Intermediate Macroeconomics, International Trade, Economics of Nonprofit Organizations

#### **LANGUAGES AND SKILLS**

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**Citizenship:** Hong Kong Special Administrative Region, China

**Languages:** English (fluent), French (intermediate), Mandarin (fluent), Cantonese (native)

**Programming:** Stata, R, Matlab, Python, ArcGIS, SurveyCTO

#### **REFERENCES**

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##### **Professor Christopher Udry (Chair)**

Department of Economics  
Northwestern University  
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##### **Professor Dean Karlan**

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##### **Professor Lori Beaman**

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